

Merlin Gogolin

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(646) 204-3834

EDUCATION

Massachusetts Institute of Technology (MIT)

Cambridge MA

Candidate for Bachelor of Science in Mechanical Engineering with Concentration in Computer Science

June 2027

GPA: 5.0/5.0

- Candidate for minors in Physics and German
- Relevant Coursework: Design and Manufacturing, Dynamics and Control, Mechanics and Materials, Fundamentals of Programming, Numerical Computation, Intro to Machine Learning

Hunter College High School (HCHS)

New York, NY

Graduated with awards from the math and physics departments

May 2023

GPA: 4.0/4.0

- SAT: 1570 (790 Math, 780 English)

PROFESSIONAL EXPERIENCE

MIT DeCoDe Lab

Cambridge, MA

Undergraduate Researcher

August 2025 – Present

- Building pipeline to automatically extract programmatic sequences from CAD data to facilitate Machine Learning endeavors

Mecado (VC-Backed AI Startup)

Cambridge, MA

Machine Learning Engineer

June 2025 – Present

- Led the initiative to adapt and train a Neural Network-based architecture for semantic labeling of faces on 3D CAD parts and assemblies, including integrating a novel inference mechanism into CAD validation software
- Benchmarked SOTA 3D learning models, adjusting and editing training parameters and architecture methodologies
- Created a labeled 3D part dataset via both classical CAD production methods and parametric, reproducible designs

DESY CMS Group

Hamburg, Germany

Research Assistant

June 2024 – August 2024

- Investigating ALPs (Axion-Like Particles) and their potential signatures at the FCC-ee (Future Circular Collider—electron-positron) using Monte-Carlo simulations, ROOT graphing tools, and statistical and physical analyses
- Published 30-page paper summarizing new research in the field of BSM (Beyond the Standard Model) particle physics
- Presented a final research presentation to the global FCC LLP group; updated CERN FCCAnalyses repository

MIT Atomistic Simulation and Energy (ASE) Research Group

Cambridge MA

Undergraduate Researcher

February 2024 – May 2024

- Characterizing thermal storage materials, specifically graphite, and investigating their thermal and mechanical properties for use in an energy storage project with the aim of storing energy as heat in graphite blocks
- Performing mechanical tests before and after heating up graphite samples to discover the effect of extreme heat on the thermal conductivity and mechanical properties of cheap graphite

MIT Scheller Teacher Education Program (STEP) Lab

Cambridge MA

Undergraduate Researcher

January 2024 – February 2024

- Developed 40-page high school chemistry curriculum on reaction rates, with 10-page extensions for equilibrium reactions, catalysts, and activation energy simulations

LEADERSHIP

MIT Teaching Assistant

Cambridge MA

Physics: Mechanics; Physics: Electricity and Magnetism; Electronics for Mechanical Systems

September 2024 – Present

- Teaching problem-solving strategies and helping students work through group problems and problem sets to develop a fundamental understanding of physics, from Newton's Laws to Maxwell's Equations
- Assisting with the building and debugging of electronic circuits across various projects, guiding students' electronics intuition

MIT Varsity Basketball

Cambridge MA

Freshman, Sophomore Starter

October 2023 – Present

- NEWMAC Offensive and Defensive Player of the Week awards sophomore year
- 2 year High School captain; led NYC division in points and rebounds, scoring over 1000 points

SKILLS & PROJECTS

Languages: German (advanced), Spanish (intermediate)

Programming: Python (+ PyTorch, Scikit, Numpy, Pandas, ROOT), C/C++, Java, MATLAB, LaTeX, Swift, Arduino, Scheme

Software: CAD (SolidWorks, Onshape, Fusion), Ansys Suite (Mechanical, Discovery, SpaceClaim, Workbench), Excel, PowerPoint

Projects and Publications:

- Held the highest score in robotics completion for historic MIT Design and Manufacturing class (2.007)
- Won first place in line-following robot competition in Electronics for Mechanical Systems class of about 100 students, breaking the previous class record; record still held
- MIT Sandbox funded full-stack app to intelligently recommend news articles to users, utilizing SOTA ML techniques
- Coding projects: created an IDE from the ground up for Scheme coding language; coded an autocorrector to intelligently guess similar words and phrases; built ML-based fantasy football mock draft simulator and player score predictor
- 30-page physics publication: "Simulating and Investigating Long-Lived Axion-Like Particles at the Proposed FCC-ee": indico.desy.de/event/46018/contributions/173511/attachments/91780/124063/Merlin_Summer_Report_ALPs_at_the_FCCee